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AstroCLIMB Shared Task

Multimodal Feature Integration for Pairwise Relationship Classification in Astronomy Literatures

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Task and Motivation

Scientific figures and captions convey results that conventional metadata may miss. AstroCLIMB asks whether the relationship between two objects can be inferred from their content.

Inputs: text-text, text-image, or image-image pairs.

Relation	Meaning
Same figure	A figure and its corresponding caption
Same paper	Objects from the same source paper
Related papers	Source papers linked by a citation
Unrelated papers	None of the preceding relations

Training data: 10,000 labeled pairs. Class counts are 1,000 same-figure pairs and 3,000 pairs for each remaining class.

Evaluation: 5-fold stratified cross-validation (seed 42).

Pairwise Feature Representation

Feature family	Count
Textual and lexical features	74
TF-IDF and LSA similarities	50
SBERT embedding interactions	96
CLIP multimodal features	60
Visual geometry and color	20
Pair and graph features	13
Total	**313**

Features include lexical overlap, shared astronomical entities and instruments, numerical overlap, structural statistics, and modality-conditioned interactions.

Model Comparison

Configuration	Macro-F1
LightGBM (mean fold score)	0.48358
XGBoost (mean fold score)	0.49150
CatBoost (mean fold score)	0.47694
Linear classifier (mean fold score)	0.48082
Weighted ensemble (raw OOF)	0.48846
Ensemble + decision multipliers	**0.50163**

System Architecture

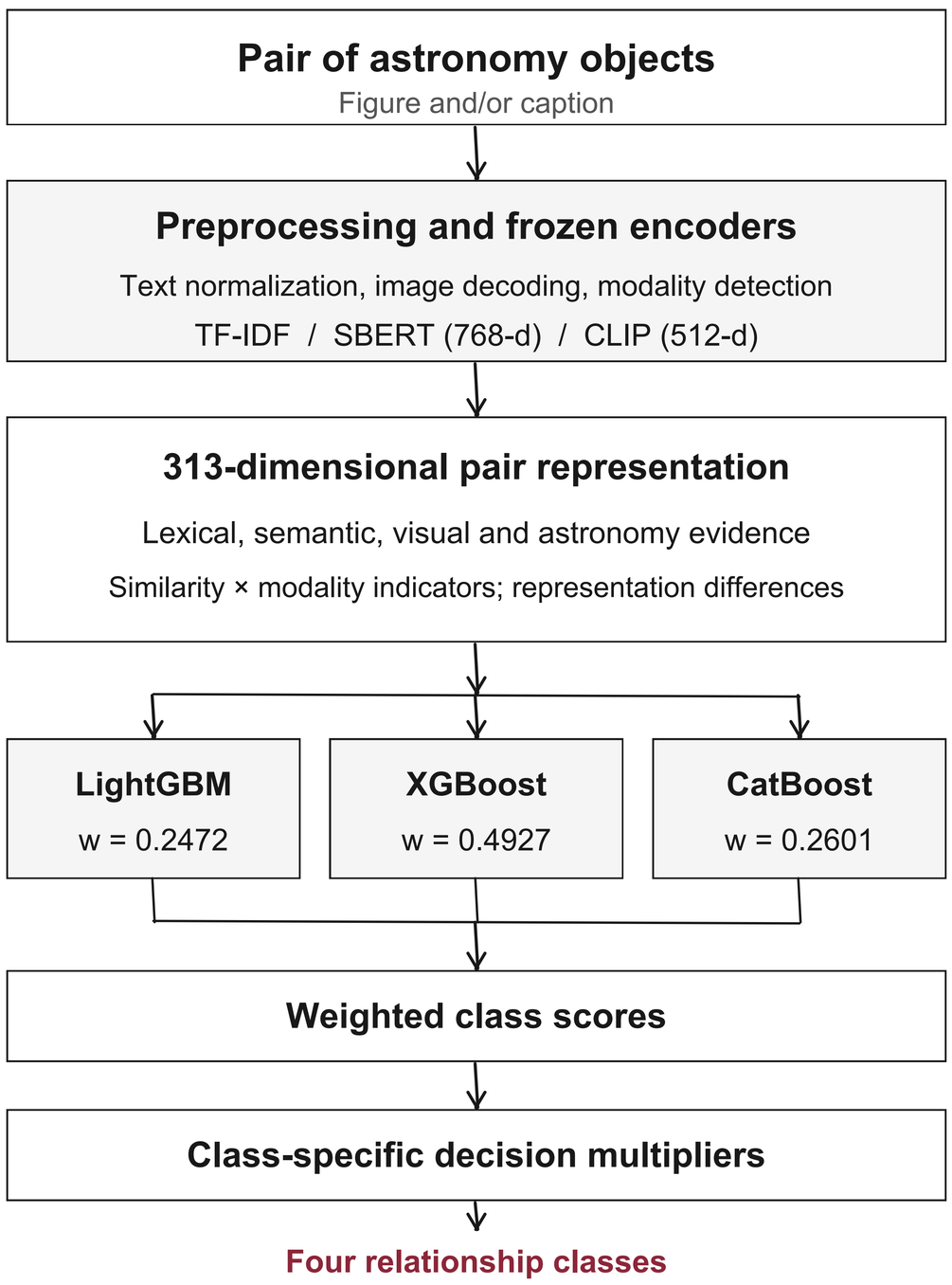


Figure 1. The final ensemble combines three tree-based models. A linear classifier is evaluated separately and receives zero ensemble weight.

Results and Error Analysis

Class	Precision	Recall	F1
Same figure	0.4612	0.7310	0.5656
Same paper	0.6123	0.4170	0.4961
Related papers	0.4071	0.4153	0.4112
Unrelated papers	0.5086	0.5613	0.5337

Related papers remain the hardest class.

Shared terminology, targets and instruments can indicate scientific similarity without establishing a citation relationship.

Kaggle score reported in the paper: 0.48121.

Rank 10 at the recorded snapshot; separate from OOF results.

Conclusions and Limitations

The ensemble reaches 0.50163 out-of-fold Macro-F1 after decision-multiplier tuning (raw: 0.48846). Multipliers use the same OOF predictions, so independent evaluation remains necessary. Controlled feature ablations, OCR-enhanced figures and richer figure-caption alignment are next steps.